Integral homology of strong Morse matching complexes of cycles

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Abstract

We determine the reduced integral homology of the full strong discrete Morse matching complex of every cycle. Apart from the five-cycle, the answer is the sum of two free abelian contributions whose degrees and ranks follow explicit formulas according to the cycle order modulo four and modulo three. The proof represents a pairing by two local order inequalities. Their conflict graph is a regular bipartite graph with a known independence complex; the strong matching complex is obtained by removing exactly two families of monochromatic vertex covers. The resulting relative chain complex splits into two copies of a simplex relative to the noncover complex of the cycle. Combinatorial Alexander duality and the long exact sequence then give the homology in all orders, extending the finite calculations of Knudson and Owens-White. The same face description gives a transfer-matrix formula for the face polynomial and a twelve-periodic reduced Euler characteristic.

Keywords. Strong discrete Morse matching; independence complex; noncover complex; combinatorial Alexander duality; cycle graph.

1 Introduction and main result

The strong discrete Morse matching complex records which local collapses can occur together in a filtration of a simplicial complex. Knudson and Owens-White introduced this complex and calculated several families [1]. For cycles, they determine the homotopy type of the pure subcomplex, but their calculation for the full complex gives homology only through cycle order 21; Remark 8.13 reports that the computation at order 22 exceeded the available memory. The present paper determines the integral homology of the full complex at every order.

Let C_n be the one-dimensional simplicial cycle on vertices v_i, with $i \in \mathbb{Z}/n\mathbb{Z}$ and $n \geq 3$. In particular, C_3 has no two-dimensional face. Write

$$K_n = \mathcal{SM}(C_n)$$

for the simplicial complex generated by the strong matchings arising from all total orders of these vertices, as in [1, Definition 4.2]. Thus every subset of a generated matching is a face. We use the full complex, which need not be pure, and adjoin the empty face for augmented chains. All homology groups in this paper have coefficients in $\mathbb{Z}$ unless stated otherwise.

Two degree functions and their multiplicities describe the answer. Set

$$q(n) = \left\lfloor \frac{2n-5}{3} \right\rfloor, \qquad d(n) = \begin{cases} n/2 - 1, & n \text{ even}, \\ (n-1)/2, & n \equiv 1 \pmod 4, \\ (n-3)/2, & n \equiv 3 \pmod 4, \end{cases} \tag{1}$$

and

$$a(n) = \begin{cases} 3, & 4 \mid n, \\ 1, & 4 \nmid n, \end{cases} \qquad b(n) = \begin{cases} 4, & 3 \mid n, \\ 2, & 3 \nmid n. \end{cases} \tag{2}$$

Theorem 1.1. *For $n \geq 3$, $n \neq 5$, and every $j \geq 0$,*

$$\widetilde{H}_j(K_n; \mathbb{Z}) \cong \mathbb{Z}^{\,a(n)\mathbf{1}_{\{j=d(n)\}} + b(n)\mathbf{1}_{\{j=q(n)\}}}. \tag{3}$$

For $n = 5$, the only nonzero reduced integral homology group is $\widetilde{H}_1(K_5; \mathbb{Z}) \cong \mathbb{Z}$. In particular, the integral homology of every K_n is torsion-free.

The two contributions in (3) occupy the same degree precisely when $n \in \{3, 4, 6, 8, 9\}$, where their ranks add. They occupy distinct degrees for every $n \geq 10$. For example, the first orders beyond the previous computation give

n	Nonzero reduced integral homology
$22, 23$	$\widetilde{H}_{10}(K_n) = \mathbb{Z}$, $\widetilde{H}_{13}(K_n) = \mathbb{Z}^2$
24	$\widetilde{H}_{11}(K_{24}) = \mathbb{Z}^3$, $\widetilde{H}_{14}(K_{24}) = \mathbb{Z}^4$
$25, 26$	$\widetilde{H}_{12}(K_n) = \mathbb{Z}$, $\widetilde{H}_{15}(K_n) = \mathbb{Z}^2$

These values follow from the theorem without constructing the complexes in those orders. The theorem determines homology; it does not assert that K_n is homotopy equivalent to the corresponding wedge of spheres.

The proof separates the local and global requirements on a vertex order. Local compatibility gives an independence complex L_n belonging to the bipartite family studied by Nilakantan and Shukla [2]. The only compatible signed sets that fail to extend to a total order are two monochromatic families of vertex covers. This face-level description identifies the relative chain complex of (L_n, K_n) with two copies of a simplex modulo the noncover complex of C_n. The cycle-independence calculation of Kozlov [3, 4] and integral combinatorial Alexander duality [5] then compute the relative groups. Their degree lies above the ambient homology degree, except at $n = 5$, so the long exact sequence determines $\widetilde{H}_*(K_n)$.

The ingredients in this calculation are established. In particular, the cycle noncover homotopy types are also included in Kim and Kim's results [6], and the ambient graphs have a circular- or Möbius-ladder description, with homotopy calculations given by Agarwal and Basak [7]. The contribution here is the exact signed-order description of K_n inside L_n and the relative-chain calculation that returns these known invariants to the strong matching complex. Section 4 derives face enumeration from the same description.

2 From vertex orders to signed faces

For a total order of the vertices, the descending link of v_i in the cycle consists of its earlier neighbors. In the strong matching construction of [1, Section 3], the vertex is paired exactly when that link is a cone. Since this link is a discrete set with at most two vertices, it is a

cone exactly when it contains one vertex. The pairing is then with the edge joining v_i to that unique earlier neighbor. This also holds for the one-dimensional triangle C_3.

Represent the two possible pairings at v_i by

$$(i,+) = \big(v_i, \{v_{i-1}, v_i\}\big), \qquad (i,-) = \big(v_i, \{v_i, v_{i+1}\}\big).$$

They are produced by a total order precisely under the respective conditions

$$(i,+): \quad v_{i-1} < v_i < v_{i+1}, \qquad (i,-): \quad v_{i+1} < v_i < v_{i-1}. \tag{4}$$

Here $<$ refers to the filtration order. Thus a selected sign prescribes the directions of both incident cycle edges, with an arrow pointing from an earlier to a later vertex. Every $+$ sign prescribes the direction of increasing cyclic index, and every $-$ sign prescribes the reverse direction.

Define a bipartite graph G_n on the $2n$ signed vertices by

$$(i,+) \sim (j,-) \quad \Longleftrightarrow \quad j \in \{i-1, i, i+1\} \pmod{n}, \tag{5}$$

and put $L_n = \mathrm{Ind}(G_n)$. We call a signed set *monochromatic* if all of its vertices have the same sign; its *support* is the set of its indices in $\mathbb{Z}/n\mathbb{Z}$.

Proposition 2.1 (Signed face description)**.** *A signed set S is a face of K_n if and only if it is independent in G_n and is not a monochromatic set whose support is a vertex cover of C_n.*

Proof. The inequalities in (4) conflict when opposite signs occur at the same index or at neighboring indices: in the latter case they give opposite directions to their common edge. Signs of the same type agree on overlapping edges, while opposite signs at nonneighboring indices involve disjoint edge sets. Consequently, independence in G_n is equivalent to consistency of all prescribed edge directions.

Assume this local consistency. A directed cycle in the partially oriented C_n must use every edge of C_n in the same direction, since the underlying graph has no other cycle. All edges are forced in the direction of increasing index exactly when the selected signs are all $+$ and their support meets every edge. The reverse direction occurs exactly for an all-$-$ vertex cover. Neither set of inequalities can hold in a total order.

In every other case the prescribed directed graph is acyclic and has a topological ordering. Such an order satisfies all inequalities required by S, so its strong matching contains every selected pairing. It can contain additional pairings, which is why the downward-closure convention in the definition of K_n is needed. Hence S is a face of K_n. □

Figure 1 illustrates why local compatibility alone is insufficient: the selected pairings can agree on every shared edge and still require an impossible cyclic order. Proposition 2.1 shows that this is the only failure. It also puts the ambient complex into a previously calculated family.

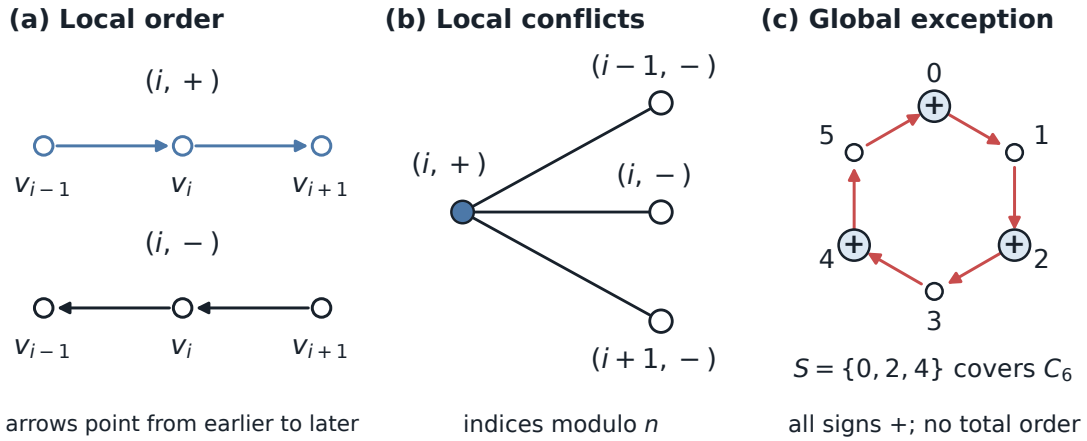


Figure 1: Local order constraints and their global exception. A sign fixes the two incident edge directions; opposite signs at equal or neighboring indices conflict. On C_6, the $+$ signs at indices $0, 2, 4$ cover every edge and force a directed cycle, so they form a face of L_6 but not of K_6. Arrows describe inequalities on the original cycle, not arrows in its Hasse diagram.

Lemma 2.2. *For every $n \geq 3$,*

$$\widetilde{H}_j(L_n;\mathbb{Z}) \cong \begin{cases} \mathbb{Z}^{a(n)}, & j = d(n), \\ 0, & j \neq d(n), \end{cases} \qquad j \geq 0. \tag{6}$$

Proof. Nilakantan and Shukla define G_m^r with shores $\{a_i\}$ and $\{b_i\}$ and neighborhoods $N(a_i) = \{b_i, b_{i+1}, \ldots, b_{i+r-1}\}$, with cyclic indices. The maps

$$(i,+) \longmapsto a_i, \qquad (j,-) \longmapsto b_{j+1}$$

identify G_n with G_n^3. For $n = 4t + \alpha$, $0 \leq \alpha \leq 3$, their Theorem 1.2 [2] gives

$$L_n \simeq \begin{cases} \bigvee_3 S^{2t-1}, & \alpha = 0, \\ S^{2t}, & \alpha = 1, 2, 3. \end{cases}$$

The dimensions and multiplicities are exactly $d(n)$ and $a(n)$. □

Remark 2.3. The graph G_n is a circular ladder $C_n \square K_2$ for even n, and the Möbius ladder on $2n$ vertices for odd n. Indeed, remove the edges joining equal indices. Advancing an index then flips the sign, giving two n-cycles when n is even and one $2n$-cycle when n is odd. Restoring the removed edges gives rungs in the first case and antipodal chords in the second. Thus the ladder calculations in [7, Theorems 12 and 15] give another route to (6).

3 The relative complex and integral homology

Proposition 2.1 locates every omitted face in one of two full simplices, namely the simplex Δ_+ on the $+$ vertices and the simplex Δ_- on the $-$ vertices. To describe the portion retained in either simplex, put $I_n = \mathrm{Ind}(C_n)$ and define its Alexander dual on the fixed n-element index set by

$$A_n = I_n^* = \{S \subseteq \mathbb{Z}/n\mathbb{Z} : (\mathbb{Z}/n\mathbb{Z}) \setminus S \notin I_n\}. \tag{7}$$

A set S belongs to A_n exactly when its complement contains an edge, or equivalently when S fails to cover that edge. Thus A_n is the noncover complex of the cycle, and

$$K_n \cap \Delta_+ \cong A_n \cong K_n \cap \Delta_-.$$

This description identifies the generators and the boundary maps of the relative chains.

Lemma 3.1. *There is an isomorphism of integral chain complexes*

$$C_*(L_n, K_n; \mathbb{Z}) \cong C_*(\Delta^{n-1}, A_n; \mathbb{Z})^{\oplus 2}. \tag{8}$$

Moreover, $H_0(L_n, K_n; \mathbb{Z}) = 0$*, and for* $k \geq 1$*,*

$$H_k(L_n, K_n; \mathbb{Z}) \cong \begin{cases} \mathbb{Z}^{b(n)}, & k = q(n) + 1, \\ 0, & k \neq q(n) + 1. \end{cases} \tag{9}$$

Proof. The relative chain group has as basis the faces of L_n that are absent from K_n. Each such face is monochromatic, and deleting a vertex cannot change its sign. The two families therefore give subcomplexes of the relative chain complex. Order the vertices in each family by their index representatives $0, \ldots, n-1$. Relabeling either family by these indices preserves every simplicial boundary sign and identifies its relative chain complex with $C_*(\Delta^{n-1}, A_n; \mathbb{Z})$. This proves (8) over the integers. Both K_n and L_n contain every signed vertex, so their relative chain group in degree zero is zero, as is their augmented relative group in degree -1.

The simplex is contractible. The long exact sequence of (Δ^{n-1}, A_n), followed by integral combinatorial Alexander duality [5, Theorem 1.1], gives, for $k \geq 1$,

$$H_k(L_n, K_n; \mathbb{Z}) \cong \widetilde{H}_{k-1}(A_n; \mathbb{Z})^{\oplus 2} \cong \widetilde{H}^{\,n-k-2}(I_n; \mathbb{Z})^{\oplus 2}. \tag{10}$$

The ground set for this duality has n elements, not $2n$: it is the index set of one sign family.

Kozlov's cycle-independence calculation [3, Proposition 5.2], also proved in [4, Proposition 11.17], is

$$I_n \simeq \begin{cases} S^{r-1} \vee S^{r-1}, & n = 3r, \\ S^{r-1}, & n = 3r + 1, \\ S^r, & n = 3r + 2. \end{cases} \tag{11}$$

Substituting these dimensions into (10) places the relative group in degree $2r - 1$ when $n = 3r$, and in degree $2r$ when $n = 3r + 1$ or $3r + 2$. These degrees equal $q(n) + 1$. The multiplicities are respectively four and two, as in $b(n)$. □

The relative calculation supplies a group one degree above $q(n)$, while Lemma 2.2 places ambient homology at $d(n)$. Their order is the final issue in the proof. Figure 2 illustrates the comparison, including the exceptional coincidence at $n = 5$.

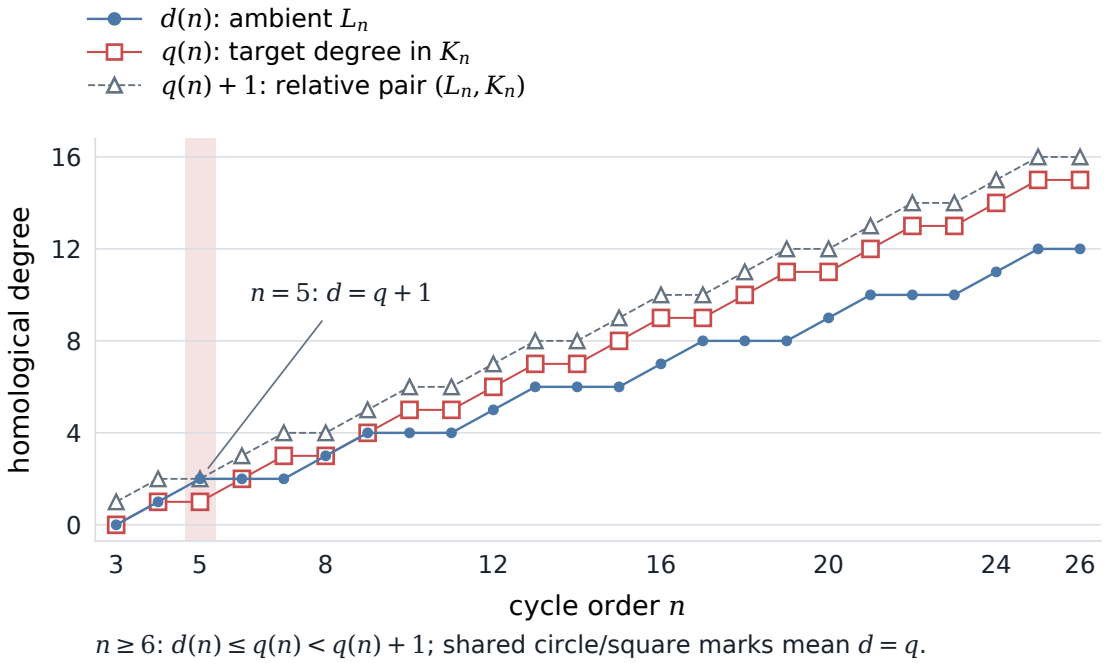


Figure 2: The homological degrees in the proof, calculated from (1) for the displayed cycle orders. Ambient homology occurs at $d(n)$, and relative homology at $q(n)+1$. The long exact sequence transfers the latter contribution to degree $q(n)$ when $q(n)+1 > d(n)$. At $n=5$ the ambient and relative degrees coincide, so this addition rule does not apply. The all-order comparison is proved in Lemma 3.2.

Lemma 3.2. *For $n \geq 6$, $q(n) \geq d(n)$. For $n \geq 3$, equality holds exactly at $n \in \{3,4,6,8,9\}$, and $q(n) < d(n)$ holds exactly at $n = 5$.*

Proof. Write $n = 12m + s$ with $m \geq 0$ and $6 \leq s \leq 17$. From (1),

$$q(n) - d(n) = 2m + q(s) - d(s).$$

The twelve base differences are

s	6	7	8	9	10	11	12	13	14	15	16	17
$q(s) - d(s)$	0	1	0	0	1	1	1	1	1	2	2	1.

They prove both claims in this range. Direct substitution gives differences $0, 0, -1$ at $n = 3, 4, 5$. □

Proof of Theorem 1.1. First suppose $n \geq 6$, and abbreviate $q = q(n)$ and $d = d(n)$. By Lemma 3.2, $q+1 > d$, so $\widetilde{H}_{q+1}(L_n) = 0$. Lemma 3.1 also gives $H_q(L_n, K_n) = 0$. The reduced long exact sequence of the pair therefore contains

$$0 \longrightarrow \mathbb{Z}^{b(n)} \longrightarrow \widetilde{H}_q(K_n) \longrightarrow \widetilde{H}_q(L_n) \longrightarrow 0. \tag{12}$$

The right-hand group is either zero or $\mathbb{Z}^{a(n)}$, according as $q > d$ or $q = d$. It is free, so (12) splits as a sequence of abelian groups. In every degree $j \notin \{q, q+1\}$, the adjacent relative terms vanish and the long exact sequence identifies $\widetilde{H}_j(K_n)$ with $\widetilde{H}_j(L_n)$. At $j = q+1$, exactness and the vanishings $H_{q+2}(L_n, K_n) = 0$ and $\widetilde{H}_{q+1}(L_n) = 0$ give $\widetilde{H}_{q+1}(K_n) = 0$. This proves (3), including torsion-freeness.

At $n = 3, 4$ we have $q = d = 0, 1$, respectively, and the same argument applies. For $n = 3$ it uses $H_0(L_3, K_3) = 0$ and gives the reduced sequence

$$0 \longrightarrow \mathbb{Z}^4 \longrightarrow \widetilde{H}_0(K_3) \longrightarrow \mathbb{Z} \longrightarrow 0.$$

Thus the respective reduced groups are $\mathbb{Z}^5$ in degrees zero and one, agreeing with [1, Proposition 8.5]. For $n = 5$, the degree order reverses: $q = 1$ and $d = 2$. In this case $K_5 \simeq S^1$ by the same proposition, giving the stated exceptional group. An independently reproducible collapse certificate is included with the supplementary code. □

The splitting in (12) is not canonical and is used only at the level of abelian groups. At $n = 5$ the map $H_2(L_5) \to H_2(L_5, K_5)$ can have nonzero image, so the general addition rule cannot be extended across this order. The cited homotopy equivalence, also verified by the collapse certificate, settles the integral cokernel; its rank alone would not establish absence of torsion.

4 Face enumeration

The signed description also counts faces without constructing a simplicial boundary matrix. Let

$$F_n(z) = \sum_{S \in K_n} z^{|S|} = 1 + \sum_{k \geq 1} f_{k-1}(K_n) z^k$$

and let $V_n(z)$ be the vertex-cover polynomial of C_n, with each cover S weighted by $z^{|S|}$.

Proposition 4.1. *For $n \geq 3$,*

$$F_n(z) = \operatorname{tr}(T(z)^n) - 2\operatorname{tr}(U(z)^n), \quad T(z) = \begin{pmatrix} 1 & z & z \\ 1 & z & 0 \\ 1 & 0 & z \end{pmatrix}, \quad U(z) = \begin{pmatrix} 0 & z \\ 1 & z \end{pmatrix}. \tag{13}$$

Consequently,

$$\widetilde{\chi}(K_n) = -2\cos(n\pi/2) - (-1)^n + 4\cos(2\pi n/3). \tag{14}$$

Proof. At index i, an independent set of G_n has one of three states: 0, +, or −. The only forbidden adjacent states are $+, -$ and $-, +$. With rows and columns ordered as $0, +, -$, the matrix $T(z)$ records permitted transitions and assigns weight z to a selected destination. A closed walk of length n therefore has weight z to the number of selected positions, and $\operatorname{tr}(T(z)^n)$ counts the faces of L_n. Positions are labeled throughout; no quotient by rotation is taken.

For a vertex cover, the two states are unselected and selected. Two adjacent unselected positions are forbidden, giving the matrix $U(z)$ and hence $V_n(z) = \operatorname{tr}(U(z)^n)$. Proposition 2.1 deletes one + copy and one − copy of every cover. These are distinct nonempty faces, which proves (13).

Finally, $T(-1)$ has eigenvalues $-1, i, -i$, while $U(-1)$ has eigenvalues $e^{2\pi i/3}$ and $e^{-2\pi i/3}$. Since $\widetilde{\chi}(K_n) = -F_n(-1)$, evaluating the traces gives (14). □

Equation (14) exhibits the twelve-periodic Euler characteristic as the combination of the modulo-four ambient contribution and the modulo-three relative contribution. The transfer-matrix method is standard; the subtraction of the two cover families follows from the specific face description proved here.

Verification and supplementary material

The supplementary package contains the manuscript source, vector figures and their generator, a Python verifier, an exact collapse certificate for K_5, and a small Lean module. The Python verifier compares the signed description with the downward closure generated by every vertex order for $3 \leq n \leq 8$. It also checks face counts and reduced boundary ranks over $\mathbb{F}_2$ through $n = 9$. These finite checks support the description; the integral conclusion follows from the written proof.

The collapse checker constructs K_5 from its 120 vertex orders and checks twenty free edge–triangle collapses followed by six leaf collapses, leaving a four-cycle. The Lean 4.35.0 module checks the universally quantified degree inequalities and their exceptional and equality cases in Lemma 3.2. It uses the bundled `omega` tactic, with no `sorry` or user-declared axioms. The signed correspondence, cited topological inputs, relative-chain decomposition and long exact sequence are not formalized in that module. Reproduction commands and the successful verification logs accompany the source in the supplementary archive `Strong_Morse_Cycles_Source_and_Verification_v1.0.zip`.

AI assistance

AI tools were used under the author's direction for mathematical exploration, literature research, proof checking, drafting, and preparation of the supplementary code and figures. The author is responsible for the mathematical claims and the final text. The scope of the Lean verification is stated above.

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